

MACHINE LEARNING PROJECT REPORT

Medical Insurance Cost Prediction

A Support Vector Regression (SVR) Approach with Log-Transformed Targets

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01 Executive Summary

This report documents the development of a machine learning model that predicts annual medical insurance charges for individuals based on demographic and lifestyle attributes. The project uses the classic 1,338-record medical insurance dataset and applies a Support Vector Regression (SVR) model with an RBF kernel to a log-transformed target, following outlier treatment and standard feature scaling. The final tuned model explains roughly 88% of the variance in charges on held-out test data, with a typical prediction error of about \$4,700.

0.880R² SCORE (TEST SET)**\$4,697**

RMSE

\$2,222

MEAN ABSOLUTE ERROR

1,337

RECORDS USED

The workflow spans data cleaning, IQR-based outlier detection and capping, exploratory analysis of cost drivers, one-hot encoding of categorical features, RBF-kernel SVR training on a log-scaled target, and packaging of the trained model into a Streamlit web application for interactive predictions.

02 Dataset Overview

The dataset contains 1,338 individual records with seven fields: age, sex, body mass index (BMI), number of children/dependents, smoking status, US census region, and the target variable, annual medical charges billed by the insurer. There were no missing values in any column. A single exact duplicate row was identified and removed, leaving 1,337 unique records for analysis and modeling.

Field	Type	Description
age	Numeric	Age of primary beneficiary (18–64)
sex	Categorical	Insurance contractor gender (male / female)
bmi	Numeric	Body mass index
children	Numeric	Number of dependents covered
smoker	Categorical	Smoking status (yes / no)
region	Categorical	US residential region (4 levels)
charges	Numeric (target)	Individual annual medical costs billed (\$)

03 Data Cleaning & Preparation

Duplicate Removal

One duplicate record (a 19-year-old male, non-smoker, from the northwest region) was found and dropped, and the index was reset.

Outlier Detection (IQR Method)

Outliers in the three continuous columns were flagged using the interquartile range (IQR) rule — any value falling outside 1.5x the IQR beyond the 25th/75th percentile bounds. No outliers were found in age. BMI had 9 values above the upper bound of 47.3. Charges had 139 values above the upper bound of roughly \$34,500 — expected, since medical costs are heavily right-skewed and dominated by smokers with high BMI.

Treatment Decision

Rather than dropping high-charge records — which are genuine, meaningful cases (largely driven by smoking) — only BMI was capped at its IQR upper bound (47.3) to reduce the influence of a small number of extreme measurements. The right skew in charges was instead addressed at the modeling stage via a $\log_{1p}$ transformation of the target.

04 Exploratory Data Analysis

Four views were used to examine how charges vary with age, BMI, smoking status, and number of children. Smoking status stands out immediately as the dominant driver of cost — smokers form a visibly separate, higher-charge band across nearly every other variable.

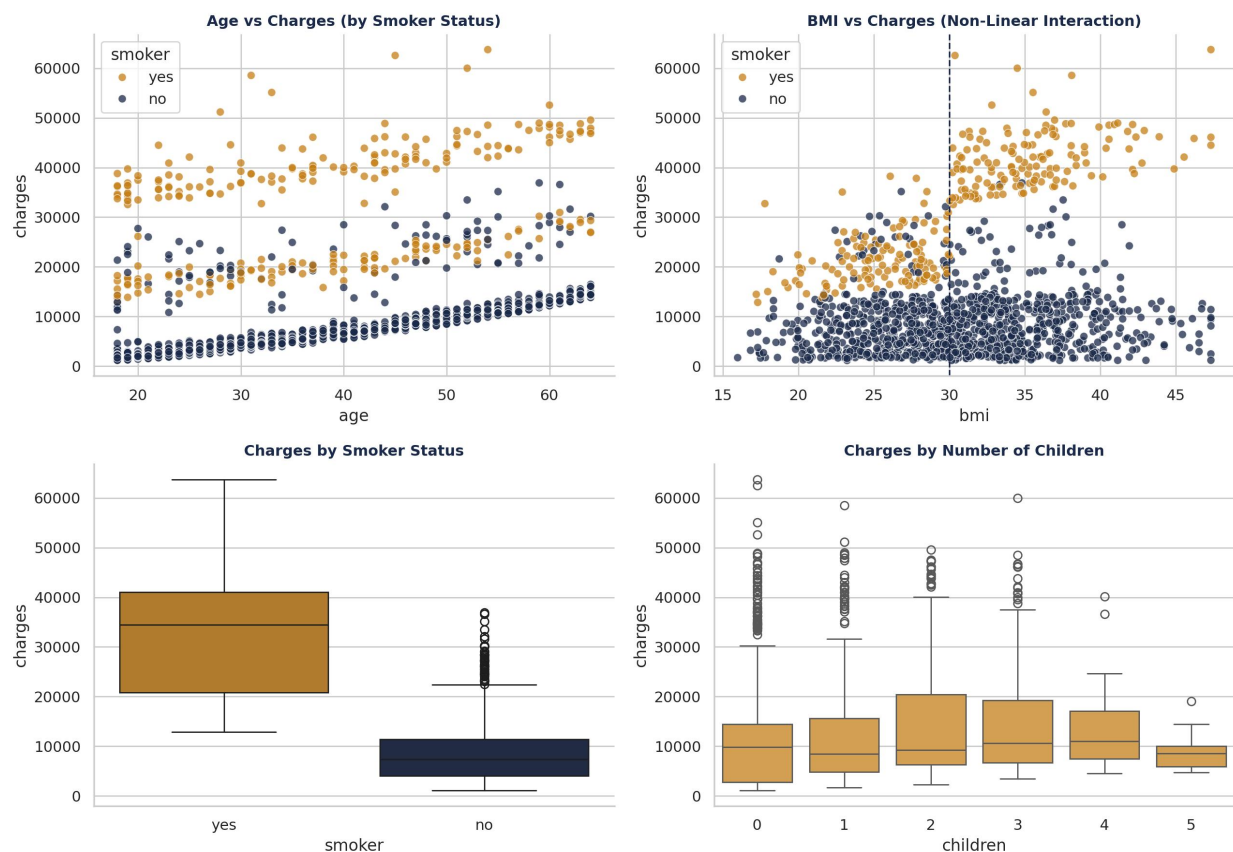


Figure 1. Charges by age, BMI, smoking status, and number of children.

- Age vs. charges: costs trend upward with age within both smoker and non-smoker groups, but smokers sit on a consistently higher, steeper band.
- BMI vs. charges: the relationship is non-linear and interacts strongly with smoking — smokers with BMI above ~30 (the clinical obesity threshold) show a sharp jump in charges, while non-smokers show only a mild BMI effect.
- Smoker status alone separates the distribution cleanly: the median charge for smokers is roughly 4x that of non-smokers, with far less overlap than any other single variable.
- Number of children shows only a weak, non-monotonic relationship with charges once other factors are accounted for.

Regional Patterns

Charges and smoking prevalence were also compared across the four US regions. The southeast shows both the highest smoker rate (25.0%) and the highest average charges (\$14,735), while the southwest has the lowest average charges (\$12,347).

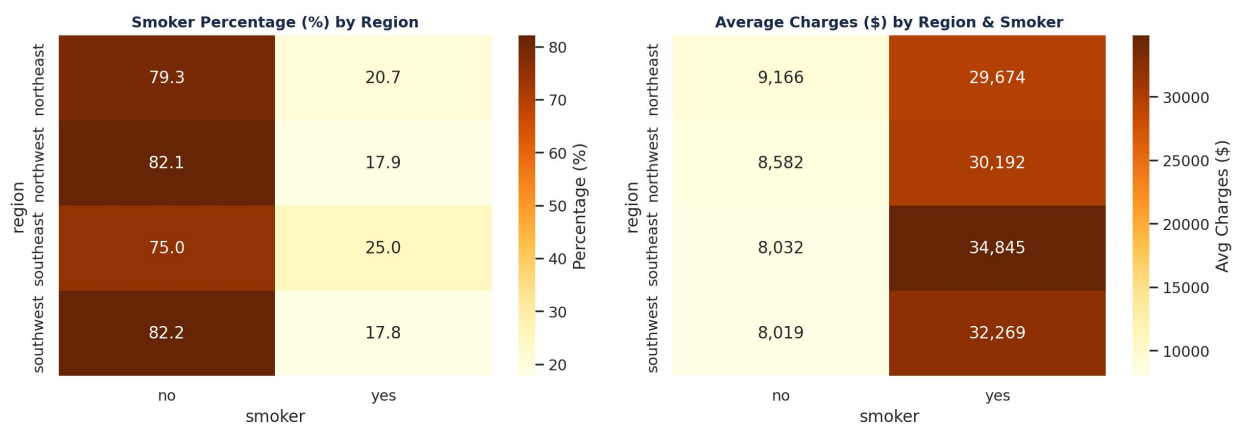


Figure 2. Smoker prevalence and average charges by region.

05 Feature Correlation Analysis

A Pearson correlation matrix over the numerically encoded features confirms the EDA findings: smoking status has by far the strongest linear correlation with charges, followed by age and BMI. Sex and region show only weak associations with cost.

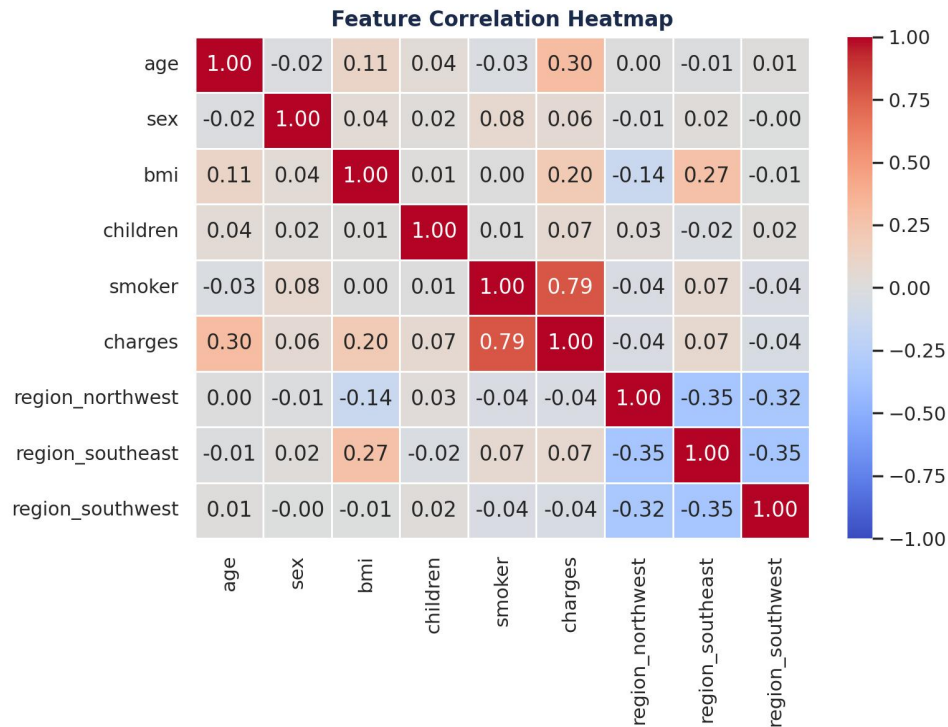


Figure 3. Correlation heatmap of encoded features.

06 Modeling Methodology

Categorical features (sex, smoker, region) were one-hot encoded with the first level dropped to avoid collinearity. The data was split 80/20 into training and test sets (random_state = 42). Because insurance charges are strongly right-skewed, the target was log-transformed (log1p) before scaling, which the notebook applies specifically to compress the influence of the small number of very high-cost cases so the regressor is not dominated by them. Both the features and the log-transformed target were standardized with separate StandardScaler instances fit only on the training data.

The regressor is a Support Vector Regression (SVR) model with an RBF kernel, configured with $C = 100$, $\gamma = 0.01$, and $\epsilon = 0.1$. Predictions are generated in the scaled log space and then reversed in two steps — inverse-scaling followed by an `expm1` transform — to return charges to their original dollar scale before evaluation.

Why SVR with a Log Target

SVR with an RBF kernel can capture the non-linear interaction between smoking status and BMI observed in the EDA (Figure 1) without manual feature engineering. Log-transforming the target keeps the small number of very high-cost, smoker-driven cases from dominating the loss function during training, while the two-step inverse transform at prediction time ensures the reported error metrics are on an interpretable dollar scale.

07 Model Performance

Evaluated on the held-out 20% test set (268 records), the tuned SVR model achieves an R^2 of 0.880, meaning it explains about 88% of the variance in actual charges. The root mean squared error is roughly \$4,697, and the mean absolute error is roughly \$2,222 — reasonable given that charges in the dataset range from about \$1,100 to over \$63,000.

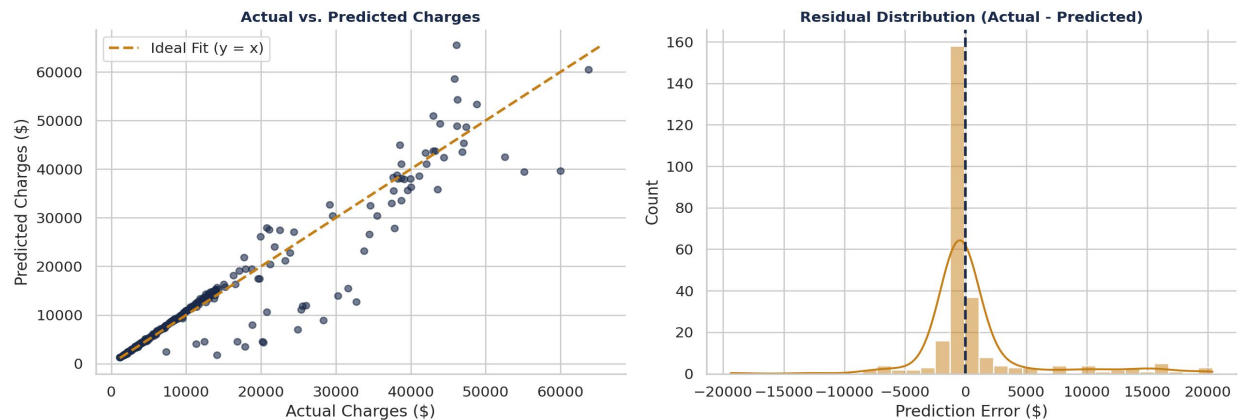


Figure 4. Actual vs. predicted charges (left) and residual distribution (right) on the test set.

The actual-vs-predicted plot shows most points tracking the ideal-fit line closely, with the model performing best on low-to-mid cost cases. The residual distribution is sharply peaked near zero, but a handful of the very highest-cost cases are under-predicted — a common pattern for right-skewed cost data, and consistent with the small cluster of points below the ideal-fit line at the top end of the actual-vs-predicted chart.

08 Hyperparameter Sensitivity

The notebook includes a comparison of four (C, gamma, epsilon) combinations intended to illustrate underfitting, a balanced default, a smooth non-linear fit, and overfitting. As run in the notebook, this comparison cell reused the pre-scaled training/test arrays from the main modeling cell but reversed only the StandardScaler step when converting predictions back to dollars — it did not also reverse the log1p transform. Every combination therefore compared log-scale predictions against dollar-scale actuals, which produced a negative R^2 (about -1.11) for all four settings and made the experiment appear to show that none of the configurations worked.

Correction Applied for This Report

Re-running the same four combinations with the full two-step inverse transform (StandardScaler inverse, then expm1) — matching the main model cell — produces sensible, differentiated results. The corrected figures below are what the experiment was designed to show.

C	Gamma	Epsilon	R^2 Score	RMSE (\$)
0.1	scale	0.1	0.759	\$6,660

10	scale	0.1	0.852	\$5,223
100	0.01	0.1	0.880	\$4,697
100	1.0	0.1	-5.282	\$33,974

The corrected results confirm the intended story: $C = 0.1$ underfits, $C = 10$ with the default gamma is a reasonable middle ground, $C = 100$ with gamma = 0.01 is the best-performing and final chosen configuration, and $C = 100$ with gamma = 1.0 overfits so severely that it performs far worse than a naive mean predictor on the test set.

09 Deployment

The final trained SVR model and its two scalers were serialized with joblib (svr_model.pkl, scaler_X.pkl, scaler_y.pkl) alongside the ordered list of model input columns (model_columns.pkl). These artifacts power a Streamlit application (app.py) that collects age, BMI, number of children, sex, smoking status, and region through simple form inputs, reconstructs the one-hot encoded feature vector, and returns a predicted annual charge after applying the same scaling and log-inverse pipeline used during training.

- Inputs: age, BMI, children, sex, smoker status, region — collected via sliders and dropdowns.
- Pipeline: input encoding → StandardScaler transform → SVR prediction → inverse scale → expm1 → predicted dollar charge.
- Dependencies are pinned in requirements.txt (streamlit, joblib, scikit-learn, pandas, numpy) for reproducible deployment.

10 Limitations & Recommendations

- Under-prediction at the extreme high end: the very highest-cost cases (typically smokers with high BMI) are somewhat under-predicted; a quantile loss or a two-stage model (classify high-risk, then regress within group) could help.
- Small, single-country dataset: 1,337 records from a US-only sample limits generalizability; charges structures vary substantially by country and insurance system.
- No formal hyperparameter search: the final (C , gamma, epsilon) values came from a manual four-point comparison rather than a grid or random search with cross-validation; a proper search would likely refine performance further.
- Fix the inverse-transform bug in the experiment cell (add the missing `np.expm1` step) before reusing that code elsewhere, since it silently produced misleading negative- R^2 results in the original notebook.
- Add cross-validation and a held-out validation set distinct from the test set to reduce the risk of the reported metrics reflecting a lucky train/test split.